IntelQCopter

Autonomous Quadcopter

TECHNICAL DESIGN AND VALIDATION REPORT



AI CONCEPT ILLUSTRATION / RED AND WHITE F450 / NOT A HARDWARE TEST PHOTO

F450	1 kHz	RP2350	ROS 2
X FRAME	FC TARGET	FLIGHT CONTROLLER	PI / JETSON TBD

RP2350 aircraft + RP2040 handheld / LoRa + reserved CC1101 links.

ROS 2 onboard computer: seeking a Raspberry Pi or NVIDIA Jetson.

Models, sensor fusion and navigation design; flight acceptance pending.

IQC-PPP-002 / 27 OCTOBER 2026 / UNIVERSITY SUBMISSION DRAFT

Student / ID / institution / supervisor / academic year: complete before submission.

Proposed targets and historical results are distinguished. Motor outputs remain disabled.

github.com/BenMohStem/IntelQCopter

Project brief and reading map

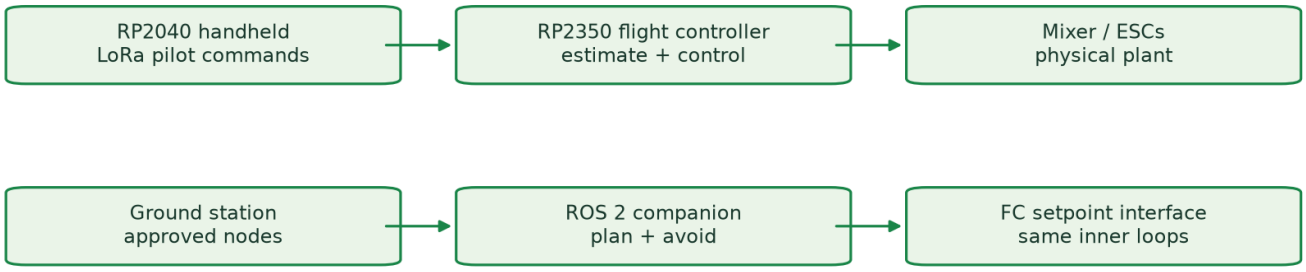
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Aim: standalone stable flight, then onboard waypoint navigation and obstacle avoidance. Target: RP2350 aircraft + RP2040 handheld. ESP32-S3/Xbox remains a separate experimental version.

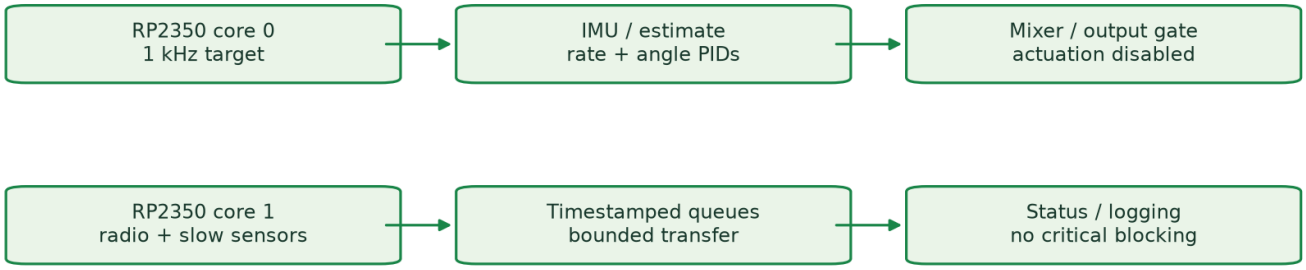
Evidence legend: historical log/replay = recorded source; analytic = explanatory model; proposed = unverified target. No new flight or Gazebo campaign was run for this report. Motor outputs remain disabled.

Math notation: bold = vector/matrix; dot = time derivative; hat = estimate; bar = sample mean; superscript T = transpose; -1 = inverse; || || = Euclidean norm; Σ = summation when used with indices; exp = exponential. Subscripts name an axis, sample or quantity. Symbols are local to each page.

System architecture and radio links



PROPOSED ARCHITECTURE — Two command paths; motor commands belong only to the FC. The companion is optional.



PROPOSED ARCHITECTURE — Proposed target partition. Existing Pico scaffold is 500 Hz with stubbed hardware; RP2350 timing remains unverified.

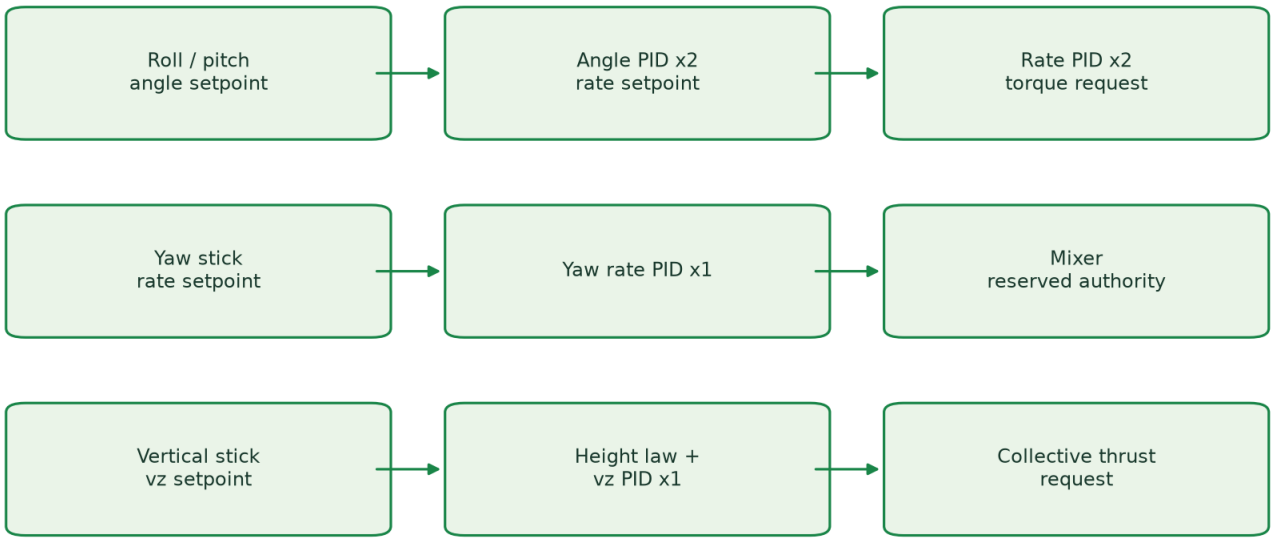
Aircraft: RP2350/Pico SDK; one flight_controller.c. Proposed core 0: bounded 1 kHz acquisition, estimation and PID; core 1: radio and slow work. Timestamped queues, one bus owner, bounded waits.

Handheld: RP2040/Pico SDK; one handler_controller.c; two centred joysticks, LB/RB, LT/RT, buttons and screen. Existing 74HC4051 acquisition needs settling/calibration tests. Display, power and gestures remain TBD.

Radio: 1 LoRa per side for pilot commands; 2 CC1101 per side reserved, not validated redundancy. Version, sequence, CRC and command age are mandatory. Airtime determines radio cadence; it is independent of the 1 kHz control loop.

Implementation gap [P6]: inspected Pico scaffold is 500 Hz with sensor/radio/output stubs. ESP32 timing cannot validate the RP2350 target. Companion: seeking Raspberry Pi 4/5 or NVIDIA Jetson; neither selected.

Flight behaviour and control architecture



PROPOSED ARCHITECTURE — Six full PID blocks are requested. The outer height-to-speed law remains a design choice.

$$e_k = r_k - y_k, \qquad u_k = K_p e_k + K_i \sum_{j=0}^{k} e_j \Delta t_j + K_d \dot{e}_{f,k}$$

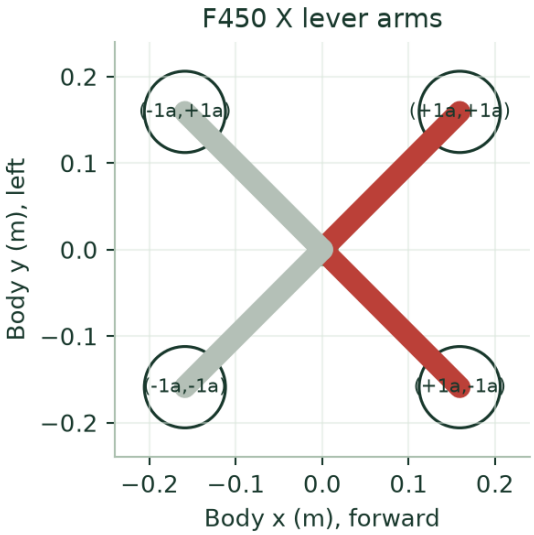
Symbols: k, j: sample indices; Δt: sample interval (s); r: reference; y: measured output; e: error; u: actuator request; Kp, Ki, Kd: proportional, integral and derivative gains (units depend on the loop); dotted e_f: filtered error derivative.

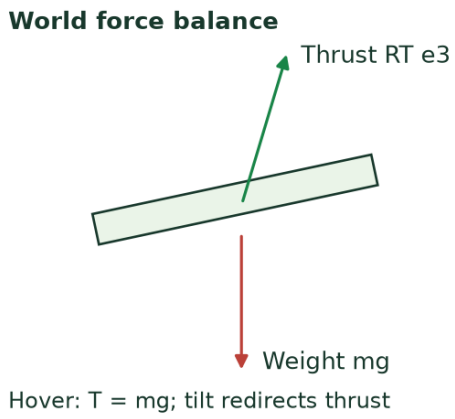
Pilot input	Controlled variable	Neutral behaviour
Right stick	Roll and pitch angles	Level attitude; no horizontal brake or position hold
Left horizontal	Yaw rate	Zero commanded yaw rate; no absolute-heading hold
Left vertical	Vertical speed	Height retention through the vertical law
RT / LT	Bounded increase / decrease	Gentle profile; reserve correction authority

Six PID blocks: 2 angle + 2 rate + 1 yaw-rate + 1 vertical-speed. An outer height-to-speed P law remains an option. Validate transitions, boost and saturation.

Why PID? I chose PID to apply what I learned: proportional response, integral error rejection, derivative damping, stability and sampling. Measurable gains connect theory to experiments. Cascades, filters and saturation still require validation.

F450 dynamics and operating point





ANALYTIC GEOMETRY — Frame convention is explicit; this drawing does not assign motor numbers or spin signs.

$$L = 0.225\ \text{m}, \quad a = L/\sqrt{2} = 0.1591\ \text{m}, \quad T = \sum_{i=1}^{4} T_i$$

$$m\dot{\mathbf{v}} = R\mathbf{e}_3 T - mg\mathbf{e}_3 - D\mathbf{v} + \mathbf{f}_{ext}$$

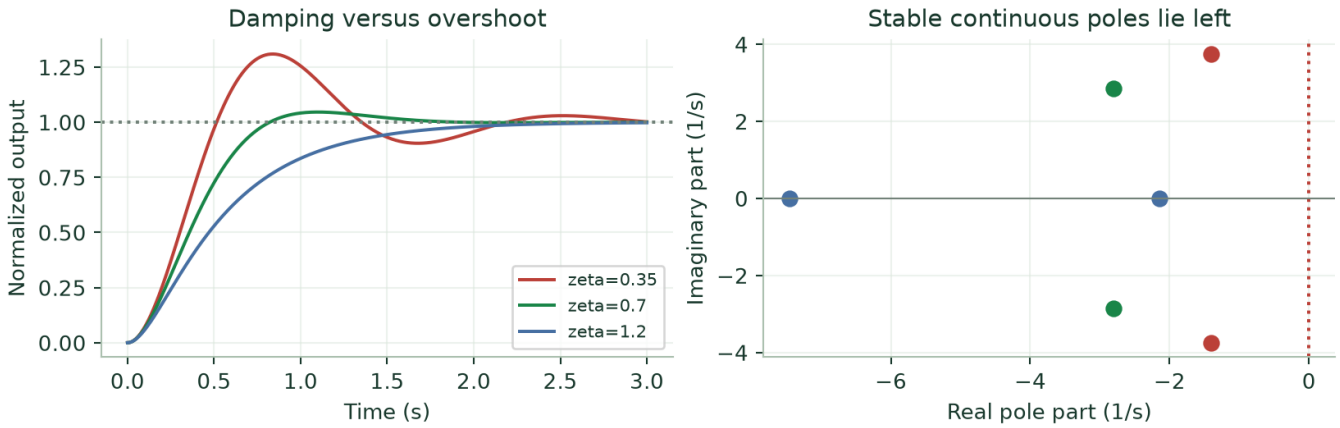
$$J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (J\boldsymbol{\omega}), \quad \boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i$$

Symbols: L: centre-to-motor distance; a: X-frame arm projection (m); i: rotor index; T_i, T: individual/total thrust (N); m: mass (kg); v: world velocity (m/s); R: body-to-world rotation; e3: upward unit vector; g: gravity (m/s²); D: drag matrix; f_ext: external force (N); J: inertia matrix (kg m²); ω: body angular velocity (rad/s); τ: torque (N m); r_i: rotor position (m); F_i: rotor force (N). Jxx/Jyy/Jzz are principal-axis inertia entries.

Supplied simulation parameter	Value	Status
Mass; stated standard deviation	1.038 kg; 0.0160312 kg	Not independently measured
Jxx = Jyy; Jzz	0.006421; 0.011853 kg m squared	Approximate COM inertias
Hover thrust, total / rotor	10.179 / 2.545 N	From mg, equal rotor loading
Motors / props / ESCs	1400 KV / 9047 / 40 A	Exact motor response unknown

Interpretation: net force accelerates the mass; lever arms create torque. Nominal ENU world/FLU body frames require physical sign checks. Reweigh the complete target assembly and identify thrust, lag and torque before numerical gain acceptance. Battery mass estimates conflict: 320 g parameter file versus old rough 150 g.

From hover model to digital control



ILLUSTRATIVE LTI MODEL — Educational second-order responses, not tuned drone gains or simulated aircraft trials.

$$G_\phi = \frac{1}{J_{xx}s^2}, \quad G_\theta = \frac{1}{J_{yy}s^2}, \quad G_r = \frac{1}{J_{zz}s}, \quad G_z = \frac{1}{ms^2}, \quad G_{act} = \frac{1}{\tau_m s + 1}$$

$$K_p = J\omega_n^2, \quad K_d = 2\zeta\omega_n J \quad \text{(ideal torque PD example)}$$

$$\mathbf{x}_{k+1} = A_d\mathbf{x}_k + B_d\mathbf{u}_k, \quad A_d = e^{A\Delta t}, \quad B_d = \int_0^{\Delta t} e^{A\tau}B\,d\tau$$

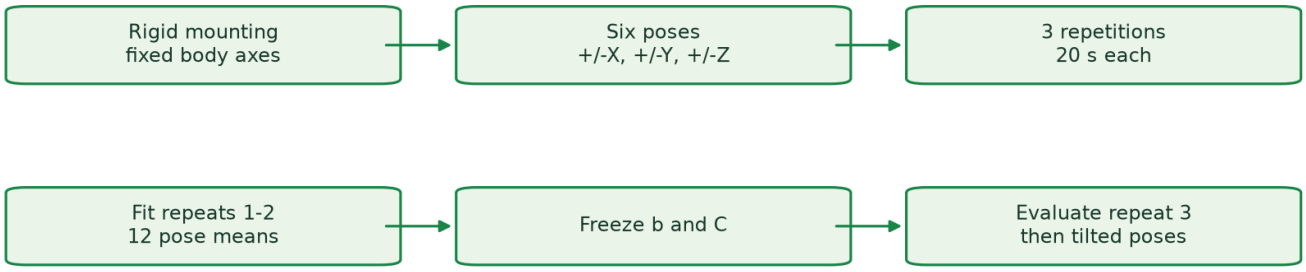
$$\phi_{k+1} = \phi_k + \Delta t\, p_k + \frac{\Delta t^2 \tau_{x,k}}{2J_{xx}}, \quad p_{k+1} = p_k + \frac{\Delta t \tau_{x,k}}{J_{xx}}$$

Symbols: G: transfer function; φ/θ: roll/pitch (rad); r: yaw rate (rad/s); z: height (m); s: Laplace variable (1/s); τ_m: actuator time constant (s); ζ: damping ratio; ω_n: natural frequency (rad/s); J: selected axis inertia. x/u: state/input vectors; A/B: continuous state/input matrices; Ad/Bd: sampled equivalents; p: roll rate (rad/s); τ_x: roll torque; τ in the integral: dummy time. Kp/Kd use torque-domain units here.

Scope: local hover LTI model; add measured actuator lag, filtering, latency, voltage and saturation. Plot = analytic damping example, not identified gains. Six final PID gains remain TBD.

Proposed targets: phase margin >=45 degrees; gain margin >=6 dB; overshoot <=10%; settling within 5% <=1 s. At 1 kHz, use measured sample intervals; sensor and ESC update rates need not equal controller rate.

Calibration by experimental design

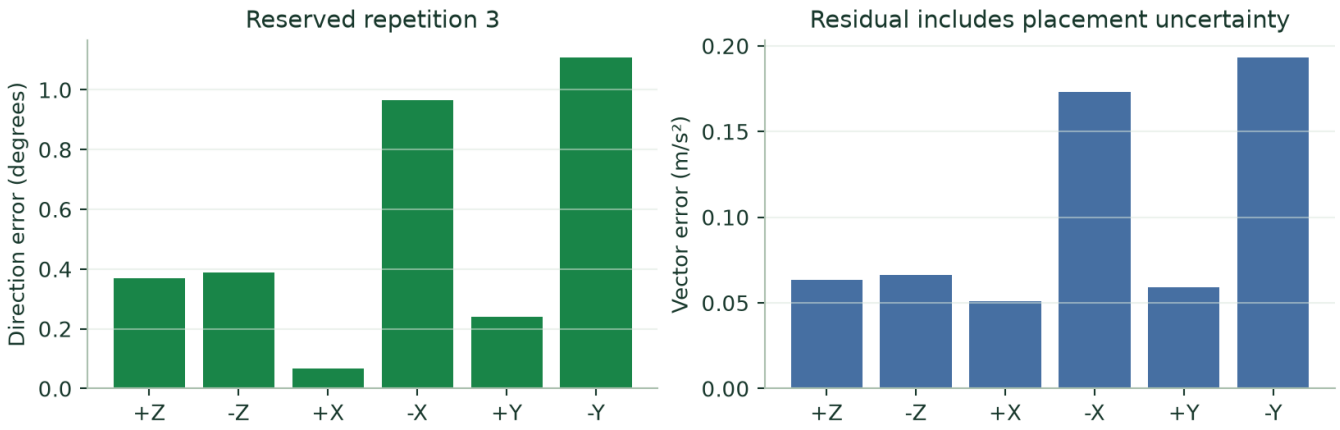


HISTORICAL TEST DESIGN — Existing 18-pose design separates fitting and evaluation. Fixture uncertainty remains.

$$\mathbf{r}_j = \mathbf{b} + A\mathbf{a}_j + \boldsymbol{\varepsilon}_j, \quad \mathbf{a}_j \in \{\pm g\mathbf{e}_x, \pm g\mathbf{e}_y, \pm g\mathbf{e}_z\}$$

$$\widehat{M} = YH^T(HH^T)^{-1} = [\widehat{\mathbf{b}}\ \widehat{A}], \quad C = \widehat{A}^{-1}, \quad \mathbf{a}_{cor} = C(\mathbf{r} - \mathbf{b})$$

Symbols: j: pose index; r: raw acceleration vector (m/s²); b: bias; A: dimensionless sensor response matrix; a_j: reference acceleration; ε: measurement error; ex/ey/ez: unit axes; Y: measured pose columns; H: reference/design columns; M: fitted bias-and-response block; C: correction matrix; a_cor: corrected acceleration. H is defined in the text below; g = 9.80665 m/s².

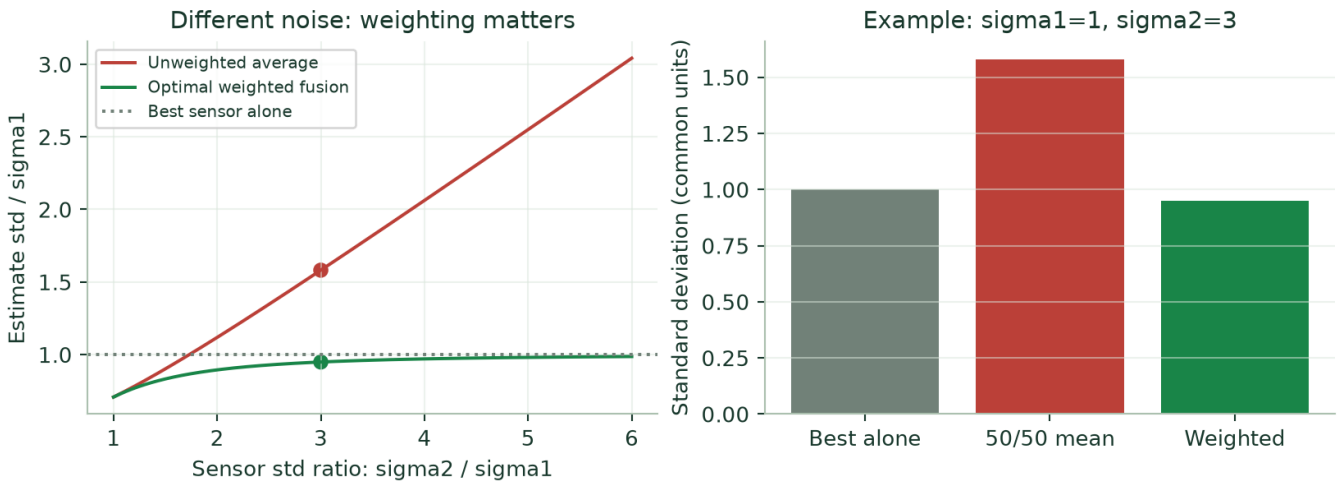


REAL LOG REPLAY SUMMARY — These residuals include reference/placement error; they are not certified sensor accuracy.

Design [P1]: 18 x 20 s recordings; repetitions 1-2 fit, repetition 3 held out. H contains [1, ax, ay, az] columns and must have full row rank. Six directions identify bias/scale/cross-axis response; one pose cannot. Historical held-out angle errors span about 0.067-1.109 degrees, including fixture uncertainty.

Gyro: motionless data estimate bias; known signed rotations test body mapping and scale. MPU offset registers are volatile: restore/read back at boot, then apply only residual software correction. Never subtract bias twice. Full accelerometer matrix correction remains software; it is not three hardware offsets. Signed gyro rotation tests remain pending.

Multisensor fusion versus a simple average



ANALYTIC INDEPENDENT-SENSOR EXAMPLE — Inverse-variance fusion improves the model variance. Correlation, bias, different units or wrong dynamics invalidate the simple comparison.

For two unbiased independent measurements of the same scalar x, let y1 = x + e1 and y2 = x + e2, with variances R1 and R2. Choose an unbiased linear estimate xhat = (1-w)y1 + w y2; its variance is minimized by inverse-variance weighting.

$$P(w) = (1-w)^2 R_1 + w^2 R_2, \quad \frac{dP}{dw} = 0 \implies w^* = \frac{R_1}{R_1 + R_2}$$

$$P^* = \frac{R_1 R_2}{R_1 + R_2} \le \min(R_1, R_2), \qquad P_{average} = \frac{R_1 + R_2}{4}$$

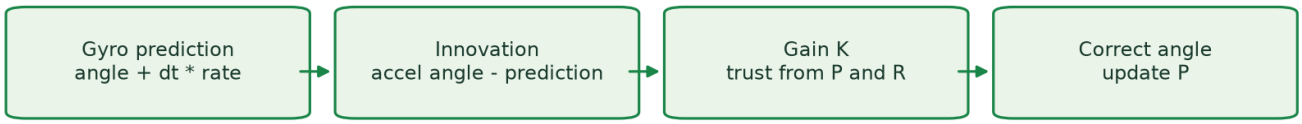
$$\mathbf{w}^* = \frac{R^{-1}\mathbf{1}}{\mathbf{1}^T R^{-1}\mathbf{1}}, \qquad P^* = (\mathbf{1}^T R^{-1}\mathbf{1})^{-1} \quad \text{(full covariance)}$$

Symbols: x: true scalar; y1/y2: observations; e1/e2: errors; R1/R2: error variances (measurement units squared); w: weight of sensor 2; P: estimate-error variance; σ: standard deviation; R in the matrix formula: full observation-error covariance; bold 1: vector of ones; bold w: weight vector; *: minimum-variance choice; P_average: equal-weight variance.

Example: sigma1 = 1 and sigma2 = 3 in the same units. A 50/50 average gives sigma = 1.581, worse than sensor 1 alone. Optimal weighting gives sigma = 0.949. The noisier sensor contributes only 10 percent. No Gaussian assumption is required for this minimum-variance linear weighting under the stated covariance model.

Kalman connection: a prediction is fused with an observation using uncertainty and dynamics. Gyro rate and acceleration cannot be averaged directly: their units and models differ. Integrate the gyro and derive a valid gravity-angle observation first. With equal independent static-sensor variance, the ordinary mean is already optimal among unbiased linear combinations. Known correlation requires the full positive-definite covariance R; bias or model mismatch can invalidate the predicted physical improvement.

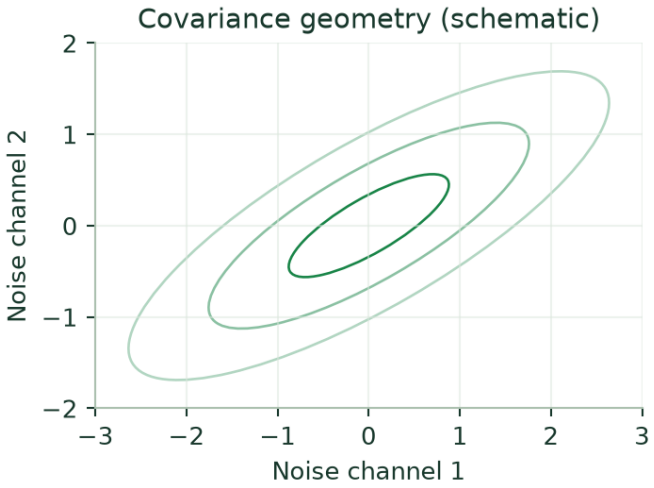
Kalman estimation and uncertainty

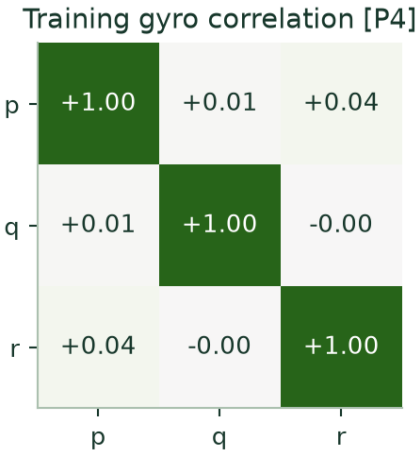


PROPOSED ARCHITECTURE — Scalar roll/pitch estimator candidate. Translational acceleration can corrupt the gravity observation.

$$\theta_k^- = \theta_{k-1} + \Delta t \omega_k, \quad P_k^- = P_{k-1} + Q_k, \quad K_k = \frac{P_k^-}{P_k^- + R_k}$$

$$\theta_k = \theta_k^- + K_k(z_k - \theta_k^-), \quad P_k = (1 - K_k)P_k^-$$





SCHEMATIC + MEASURED SUMMARY — Off-diagonal terms show shared variation. Zero covariance does not generally imply independence.

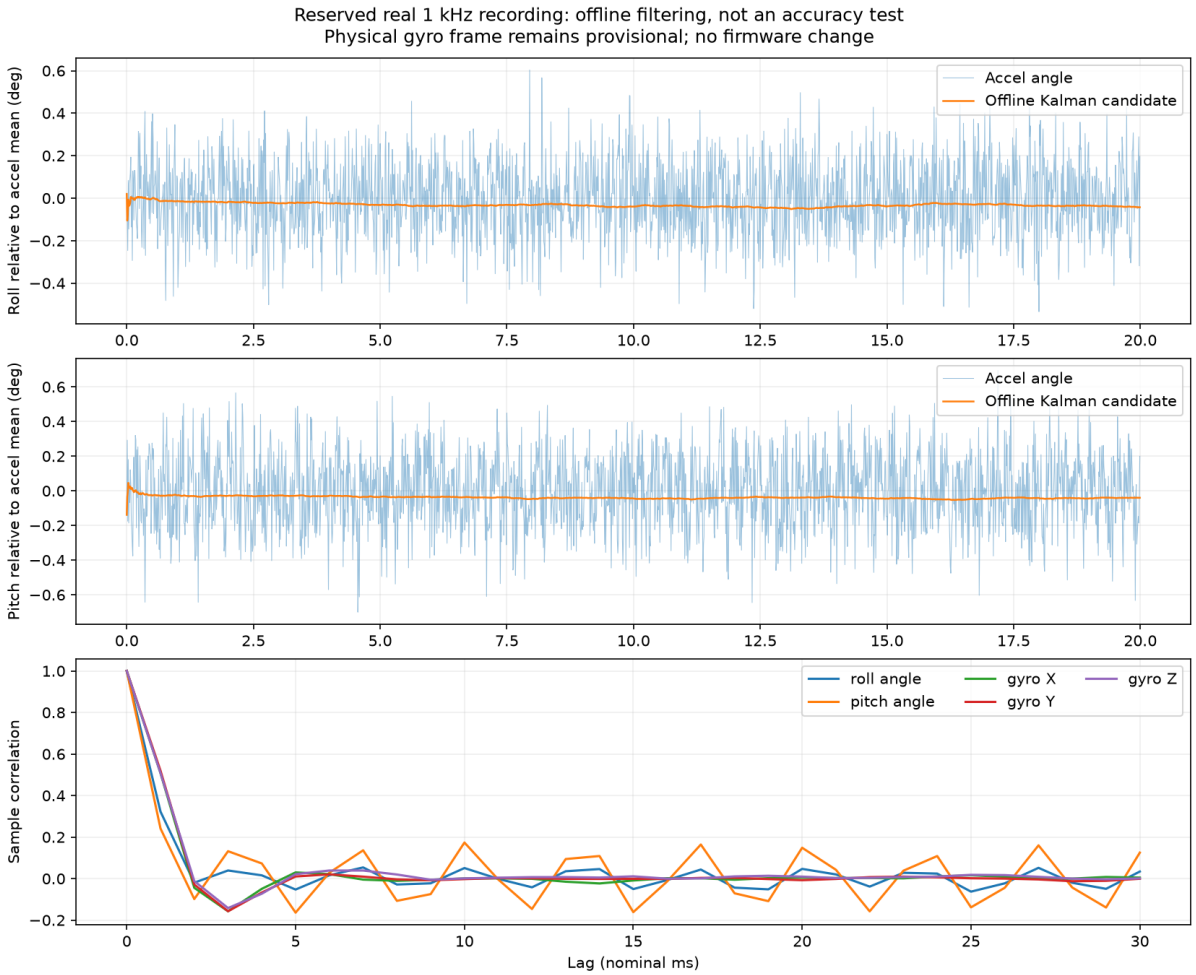
$$S = \frac{1}{N-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T, \quad \mathrm{Cov}(\bar{\mathbf{x}}) = \frac{1}{N^2}\sum_{i,j}\mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j)$$

Symbols: θ: estimated angle (rad); ω: gyro rate (rad/s); z_k: observed angle, not altitude on this page; P: angle-error variance (rad²); Q: process variance per step; R: observation variance; K: dimensionless Kalman gain; superscript minus: prediction before correction. S: sample covariance; x_i: sample vector; x-bar: mean; N: sample count; Cov: covariance operator; Σ: population covariance.

Read S: diagonal = axis variance; off-diagonal = co-variation. Only independent samples give Sigma/N. More samples do not remove bias, drift or fixture error; Gaussianity is not established.

Historical model [P4]: limiting K roll 0.00061129, pitch 0.00048326; conditional on its Q/R and timing, not flight tuning. Discrete independent rate noise gives Q = sigma_rate squared x dt squared; a continuous spectral-density model scales with dt instead. Check residuals, autocorrelation and dynamics before selecting gains.

Measured benefit and demonstrated limitation

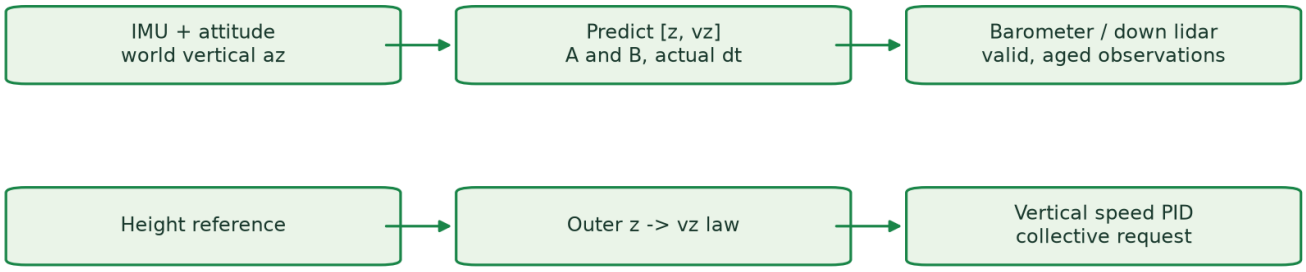


Reserved replay [P4]	Roll	Pitch
Raw angle standard deviation	0.177569 degrees	0.213074 degrees
Filter output standard deviation	0.007000 degrees	0.005692 degrees
Filter minus observation mean	-0.033769 degrees	-0.039908 degrees

Engine: original Python float64 replay of reserved real 1 kHz data; unchanged plot. Not MCU execution or flight. Fixed 2 s initialization excluded. No independent attitude reference: quieter output does not prove accuracy.

Adverse test [P4]: synthetic 0.1g lateral acceleration produced 5.191 degrees peak error; acceleration plus gyro bias produced 5.355 degrees. Apparent gravity tilt is arctan(0.1) = 5.71 degrees while magnitude is only 1.005g. Norm gating alone is insufficient. Dynamic acceptance remains blocked.

Vertical estimation and sensor geometry

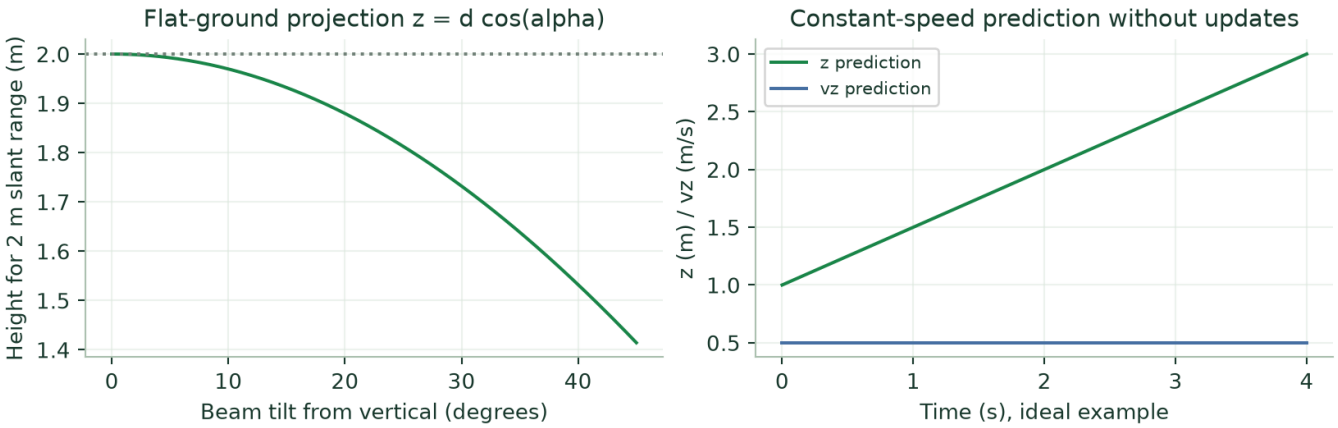


PROPOSED ARCHITECTURE — Two-state estimator and vertical control architecture proposed; barometer and lidar integration remain pending.

$$z_{k+1} = z_k + \Delta t\, v_{z,k} + \tfrac{1}{2}\Delta t^2 a_{z,k}, \qquad v_{z,k+1} = v_{z,k} + \Delta t\, a_{z,k}$$

$$P_k^- = A_d P_{k-1} A_d^T + Q_k, \quad K_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1}$$

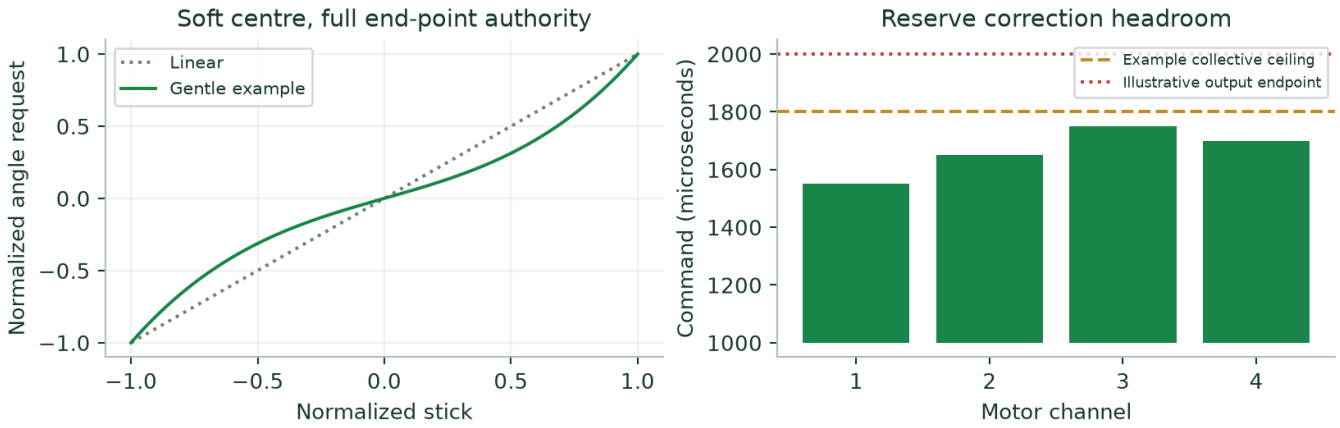
Symbols: z: height (m); v_z: vertical velocity (m/s); a_z: world-frame vertical acceleration after gravity removal (m/s²); k: sample index; Δt: interval (s); state x = [z, v_z] transpose. Ad: state transition; P: state-error covariance; Q: process covariance; H: measurement mapping; R: measurement covariance; K: Kalman gain matrix. Matrix entries carry the corresponding state/measurement units.



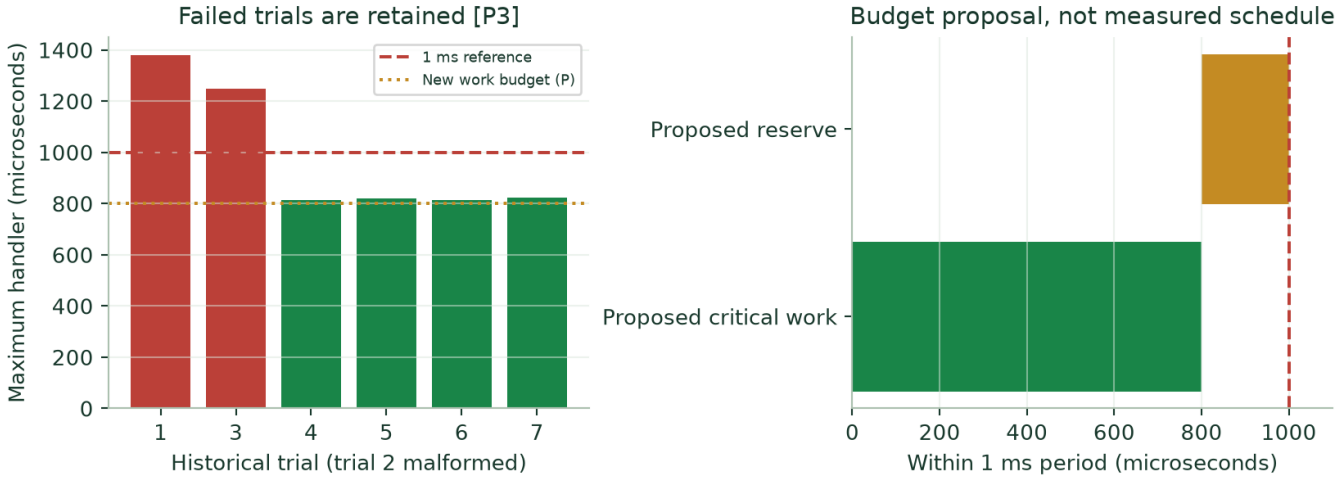
ANALYTIC EXAMPLE — A forward lidar cannot supply this height observation. Terrain and extrinsics must be checked.

The two-state model estimates z and vz; it does not automatically estimate accelerometer bias. Q must follow the selected disturbance model and sample interval. A mass variance cannot be copied directly into Q. Barometer drift, lidar status, tilt projection, terrain, measurement age and dropout all require tests. Bottom-versus-front lidar placement remains a decision, not an interchangeable software setting.

Control authority and real-time evidence



ILLUSTRATIVE CONTROL DESIGN — Actual collective/vertical law and ESC limits are TBD. A boost request is not four identical maximum outputs.



REAL LOG SUMMARY + PROPOSED BUDGET — Historical samples were not a complete PID controller; source hashes and inter-service gaps still matter.

$$u = \mathrm{clip}(u_P + u_I + u_D, u_{min}, u_{max}), \quad I_{k+1} = I_k + K_i e_k \Delta t + K_{aw}(u - u_{raw})\Delta t$$

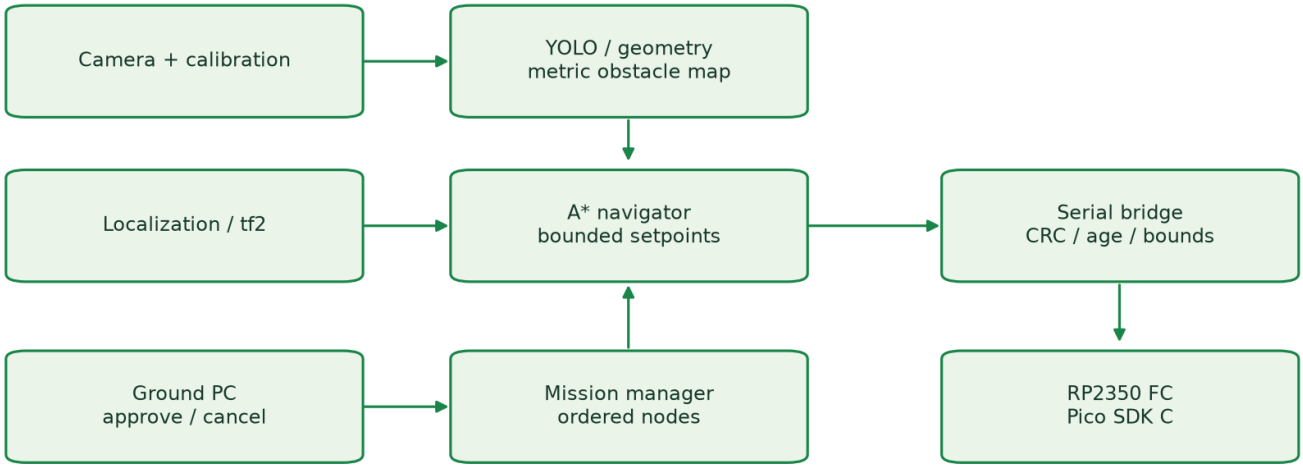
Symbols: u_raw: unsaturated request; u: limited request; uP/uI/uD: PID contributions; u_min/u_max: bounds; clip: bound to interval; I: integrator contribution (not battery current); Ki: integral gain; Kaw: anti-windup feedback gain (1/s); e: reference minus measurement; k: sample index; Δt: interval (s). Pulse widths are in microseconds.

Manual target: gentle sticks, RT increase/LT decrease; proposed collective ceiling 1700-1800 microseconds leaves PID correction headroom. This is a design target, not measured thrust. Anti-windup and transition tests precede acceptance.

Historical ESP32 timing [P3]: observed handler maxima 814, 819, 812 and 822 microseconds in successful trials; earlier failures reached 1381 and 1249 microseconds. One malformed trial was rejected. Preserve every trial. Finite acquisition-handler timing is neither complete FC timing nor a hard deadline/WCET proof, and is not RP2350 evidence.

Next gate: profile acquisition -> estimation -> PID -> output under radio/sensor load; log missed deadlines, data age and queue loss. Physical motor outputs remain disabled.

ROS 2 software implementation and interfaces



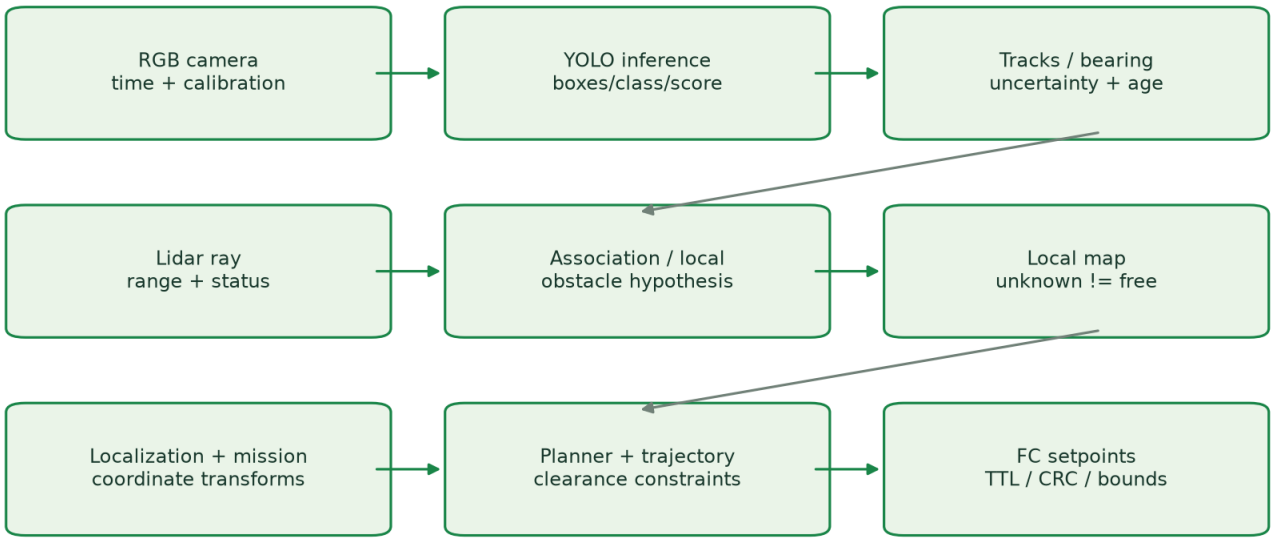
PROPOSED ROS 2 INTEGRATION — ROS 2 C++ runs on the companion. Ground-PC missions enter the mission manager; only the bounded serial interface reaches RP2350.

Boundary	Implementation contract
Ground station to companion	IP network, mission upload/approval, status and cancel; separate from LoRa pilot link
Camera and perception	Stamped RGB plus intrinsics; detections become metric obstacles only with validated geometry
Localization	Pose/velocity/covariance with coherent map, odom and body frames; freshness checked
Mission and navigator	Ordered-node progression, local-map updates, replanning and bounded references
Serial bridge to RP2350	Versioned setpoints and health; CRC, timestamps, expiry and authority arbitration
Evidence recording	rosbag2/topics, video and FC logs aligned to recorded clocks and source versions

Existing code [P6]: qcopter_autonomy contains C++ mission, navigator, YOLO and panel-pose nodes plus custom messages; qcopter_description contains the F450 URDF/RViz model. The existing mission service specifies start/finish only. A cancellable n-node ROS action with progress feedback is a proposed extension, not an already verified feature [R13].

Communication design: use small bounded histories; sensor streams prioritize freshness, mission requests need acknowledged handling, and the FC independently rejects stale commands. ROS QoS compatibility must be checked; reliable delivery alone does not guarantee a deadline [R13]. ROS 2 runs on the companion, while the RP2350 runs Pico SDK C firmware rather than a full ROS installation.

Perception for obstacle avoidance



PROPOSED ARCHITECTURE — Proposed perception-to-control pipeline. Detection is not depth, localization or a collision-free corridor.

$$\mathbf{d}_c \propto K_c^{-1}[u, v, 1]^T, \quad d_{clear} > v\tau + \frac{v^2}{2a_{brake}} + d_{margin}$$

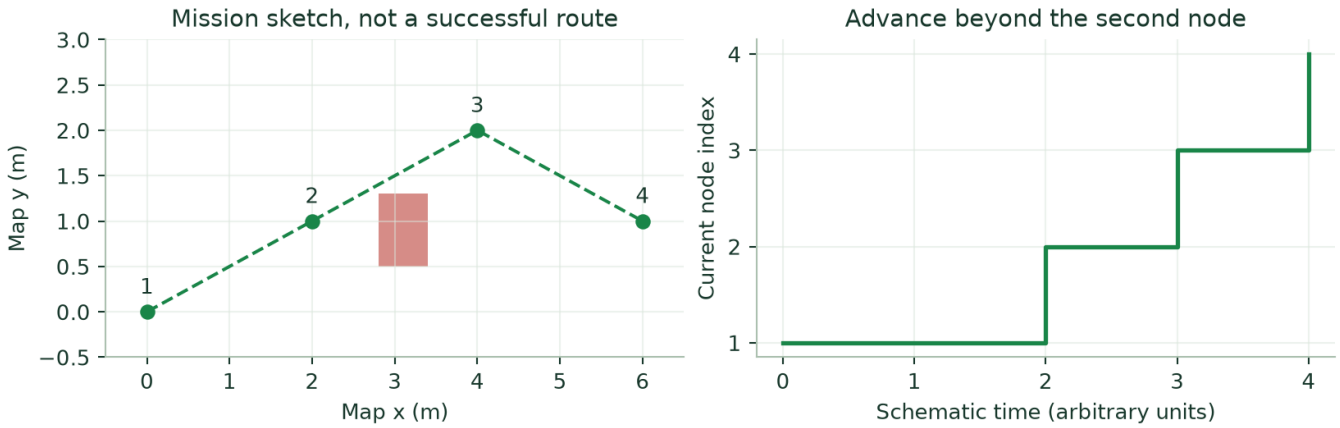
Symbols: d_c: camera-frame ray direction; K_c: intrinsic camera matrix; u/v inside [u,v,1]: pixel coordinates; d_clear: available clearance (m); v in the stopping law: speed (m/s); τ: total latency (s); a_brake: available braking deceleration (m/s²); d_margin: clearance allowance (m). These local symbols differ from control input u and world velocity vector v.

YOLO [R7]: class, box and confidence; not metric free space. Camera calibration, timestamps, tracking and camera/body transforms are required. Associate the single lidar ray only when geometry/time justify it. Unobserved space stays unknown; no detection is not proof of clearance.

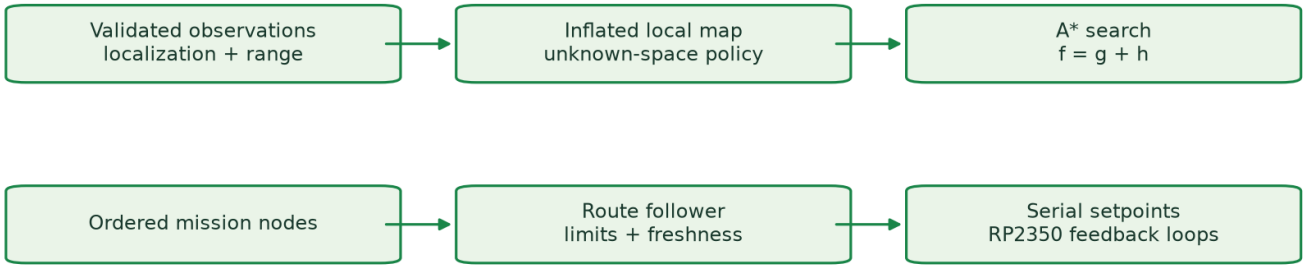
Hardware limit: one RGB camera + one fixed ray cannot observe the full collision corridor. Downward lidar aids altitude; forward lidar aids one direction. Placement remains open. Existing known-panel PnP assumes a trained detector and confirmed 1.09 x 1.69 m panel dimensions; it does not recover arbitrary obstacle depth.

Companion sought: Pi 4/5 or Jetson. Select by measured end-to-end latency, power, mass, cooling, camera/OS support and cost. Same C++ core does not imply the same binary or inference backend. Metric localization remains unresolved.

Mission nodes and online navigation



PLANNED TEST SCHEMATIC — The dashed node sequence is input geometry; it does not claim obstacle avoidance or a collision-free path.



PROPOSED ARCHITECTURE — A bounded C++ A* prototype exists; generalized n-node ROS integration, real localization and physical bridge acceptance remain pending.

$$f(n) = g(n) + h(n), \quad r_{inflate} \geq r_{vehicle} + d_{stop} + k_\sigma \sigma_{position} + d_{margin}$$

$$\|\hat{\mathbf{p}} - \mathbf{p}_i\| \leq r_{accept}, \quad \|\hat{\mathbf{v}}\| \leq v_{accept} \quad \text{for} \quad t \geq t_{dwell}$$

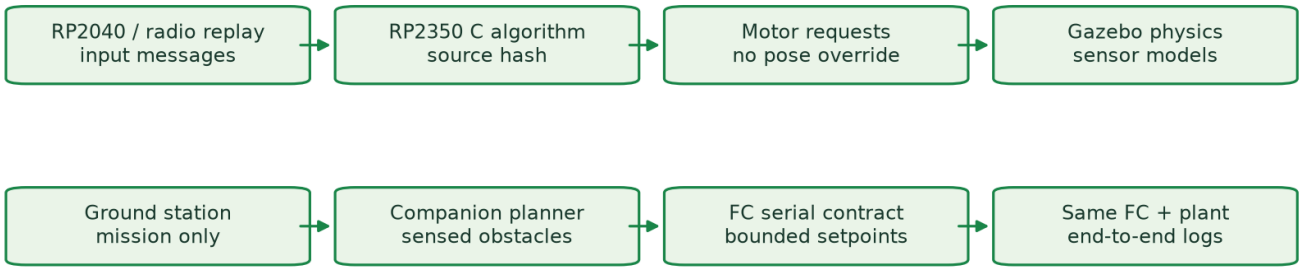
Symbols: n: search node; f: A* priority; g(n): accumulated path cost, not gravity; h(n): remaining-cost estimate; r_inflate/r_vehicle: map inflation/vehicle radius (m); d_stop/d_margin: stopping distance/margin; kσ: uncertainty multiplier; σ_position: position standard deviation (m); p-hat: estimated position; p_i: node i target; v-hat: estimated velocity; r_accept/v_accept: arrival limits; t/t_dwell: time within limits/required duration.

Target: PC chooses n nodes -> approves mission -> companion localizes, maps, replans and tracks -> FC receives bounded references. Intermediate-node advancement must be tested; final arrival and landing are separate policies.

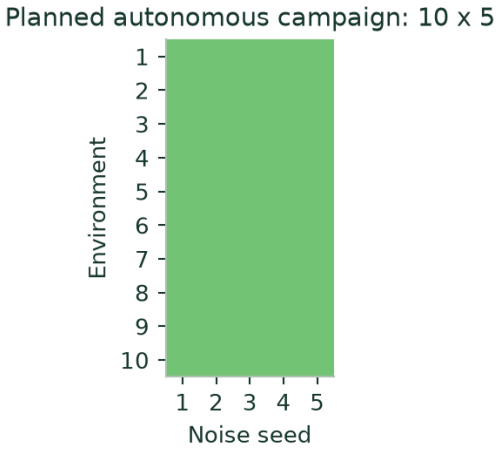
Existing [P6]: portable A* and waypoint follower; 0.2 m grid, fixed 0.65 m inflation, bounded XY map. These are prototype assumptions, not validated 3D clearance. Current ROS service accepts start/finish only; a cancellable n-node action remains proposed. Proposed arrival: 0.5 m radius and 1 s dwell, speed bound TBD.

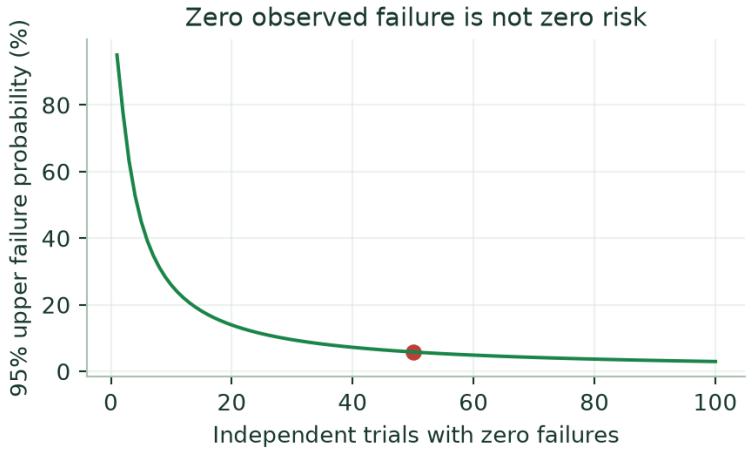
Transport: framed 3.3 V UART with CRC, sequence, units, timestamps and expiry. Proposed 10 Hz heartbeat/300 ms expiry need validation. Plugging in a companion never changes mode automatically; approved loss responses remain open.

Proving that the FC really flies the simulation



PROPOSED ARCHITECTURE — A test harness may stimulate inputs and record outputs; it may not secretly move the aircraft to pass.





PLANNED CAMPAIGN + STATISTICAL LAW — No cells represent completed missions. Independence and comparable trials are assumptions, not guarantees.

Test contract: stimulate normal handheld/radio or companion interfaces; log source hashes, messages, estimates and motor requests. Do not apply hidden pose, speed or force corrections to complete a mission. Host-compiled RP2350 C control code requires disclosed shims; it is not RP2350 instruction emulation or hardware timing evidence.

Planned Gazebo campaign: ten environments, five noise seeds each; five additional manual command scenarios. Preserve all failures. Video reviewed at 10-15 fps supports trajectory analysis; full-rate logs establish controller timing. No new campaign has been run for this report.

P targets: attitude RMS <=2 degrees, altitude RMS <=0.20 m, endpoint error <=0.5 m, zero collision in the declared campaign. They are reviewable objectives, not achievements. Zero failures in 50 independent comparable trials gives a 95 percent upper failure probability of about 5.8 percent, not zero risk.

Materials and resource budget

Subsystem	Intended materials	Selection status
Airframe and propulsion	F450 red/white; 4 x 1400 KV motors; 4 x 9047 props; 4 x 40 A ESC	Exact motor/ESC models and dynamics to verify
Power	3S 4200 mAh LiPo, marked 120C; verified avionics regulator; voltage/current sensing	Actual capacity, sag, mass and current limits to measure
Aircraft control	RP2350 board, MPU6050, BMP280, GY-271 magnetometer, TOFSense F2 mini	Exact boards/compass silicon and drivers to confirm
Handheld	RP2040 board; 2 sticks; 2 triggers; buttons; display; power; 74HC4051	Display/power/mux validation open
Radio totals	2 LoRa; 4 CC1101; 6 matched antennas	Exact modules/band/antennas open; CC1101 reserved
Autonomy option	Pi 4/5 OR Jetson, RGB camera, storage, regulator, cooling, mounts	Being sought; not purchased/selected
Integration and measurement	PCB/headers/wire; multimeter; scale; level/fixtures; reference instruments	Full 37-entry inventory in CSV

$$C_{base} = \sum_{i} n_i c_i + C_{shipping} + C_{fabrication}$$

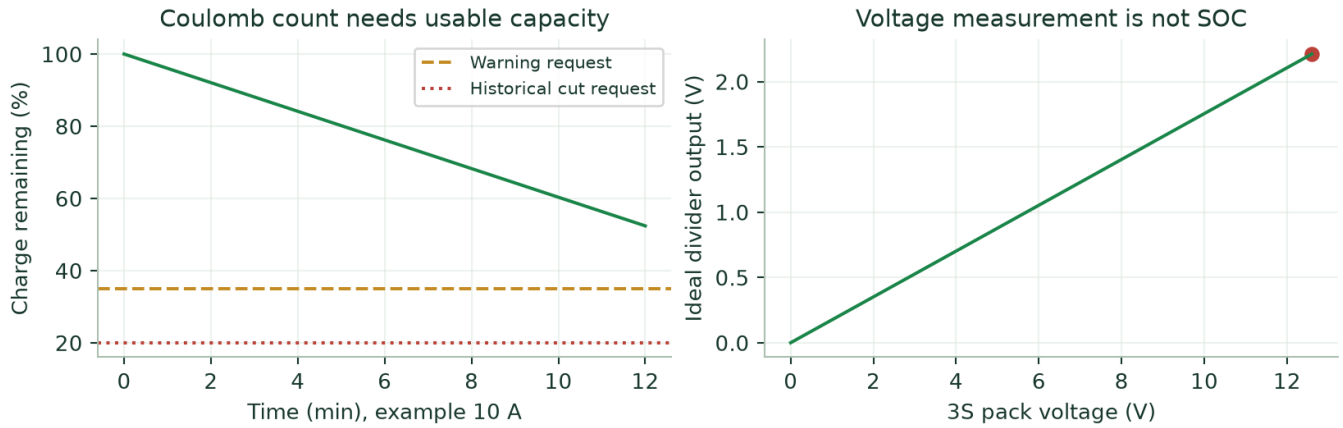
Symbols: C_base: base project cost; i: inventory item; n_i: quantity; c_i: unit price; C_shipping/C_fabrication: shipping/fabrication costs. All cost terms use one stated currency; no numeric total exists until quotations are entered.

Budget: no fabricated prices. Dated quotes, owned/to-buy status, shipping and measured mass belong in bill-of-materials.csv. Shared tools and optional companion are costed separately.

Mass update: replace prototype FC mass; add radios, antennas, camera, computer, regulator and mounts. Recompute centre of mass and inertia before gain validation.

Software: Pico SDK C; ROS 2 C++; Gazebo; RViz; YOLO; Python analysis; Git/GitHub. ESP32/Arduino/Xbox belongs to the alternative prototype. Isaac Sim excluded.

Battery estimation and state policy



ANALYTIC EXAMPLE — This is not measured endurance. Whole-pack monitoring cannot protect each cell; a motor cut in flight causes a fall.

$$SOC(t) = SOC(0) - \frac{1}{C_{usable}} \int_0^t I(\tau)\, d\tau, \qquad V_{ADC} = V_{pack} \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

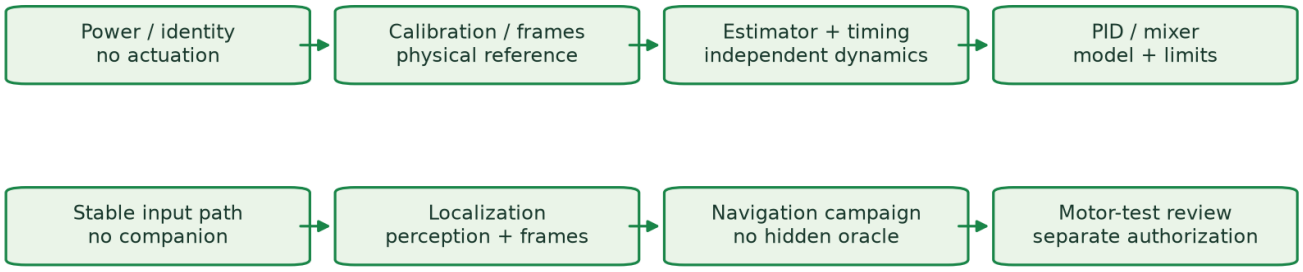
Symbols: SOC: remaining usable-charge fraction; SOC(0): initial fraction; t/τ: elapsed/dummy time (s); I: discharge current (A, positive out); C_usable: usable charge (A s; multiply Ah by 3600); V_ADC/V_pack: divider output/pack voltage (V); R_top/R_bottom: upper/lower divider resistances (ohms).

Measurements: calibrated whole-pack voltage and current, useful capacity, voltage sag and sensor validity. The plots show ideal laws, not measured endurance or an SOC calibration. Marked 120C and four 40 A ESCs do not establish a usable total discharge current.

Historical requested thresholds: warning at 35 percent and motor disarm at 20 percent, without automatic safety landing. A motor cut in flight causes a fall; reconfirm this policy explicitly before any flight phase. Hysteresis, persistence, uncertainty and fault responses remain TBD. Whole-pack software monitoring does not protect individual cells.

Current states: BOOT -> SELF_TEST -> BENCH_READY only. Reset does not arm. Physical motor outputs remain disabled. One verified regulated avionics source; common grounds; no parallel BEC positives; propulsion current stays outside FC copper. Separate authorization is required for motor tests.

Verification gates and professional method



PROPOSED ARCHITECTURE — Offline preparation is allowed; failed earlier gates invalidate affected downstream acceptance.

Gate	Required evidence before progression
Bench integrity	Common ground, regulated supplies, non-actuating outputs, reset/readback
Sensors and calibration	Held-out poses, signed gyro rotations, reference uncertainty, invalid/dropout recovery
Real-time control	Full-load deadlines, stale-command rejection, PID saturation, headroom, source-path trace
Pilot and radio	RP2040 inputs/gestures; LoRa airtime, loss, reconnect; coexistence
Autonomous mission	Localization, every intermediate node, unknown obstacles, sensor loss; no oracle
Power and flight release	Reference voltage/current/SOC tests; approved loss/battery policies; separate motor authorization

Acceptance targets, not achievements: attitude RMS <=2 degrees; height RMS <=0.20 m; endpoint <=0.5 m; zero collisions in the declared campaign. All 50 requirements and 13 verification cases remain in the repository.

Professional method: requirement -> model -> experiment -> evidence -> decision. Each checked change gets a descriptive commit and GitHub push. Preserve failed trials; repair a failed step and invalidate downstream acceptance. No history rewriting or invented pass.

Submission completion: add student/institution/supervisor, authentic personal reflection, actual contributions, library/AI attribution under university policy, dated quotations and rubric mapping. No full-mark or institutional endorsement is claimed.

Repository sources and evidence access

Public engineering repository
https://github.com/BenMohStem/IntelQCopter
Source, versions, report, measurements and scoped commit history.

Version	Role
main	Intended RP2350 FC + RP2040 handheld; normal ROS 2 packages
esp32-xbox	Alternative ESP32-S3/Xbox prototype
codex/stepwise-esp32	Incremental sensor evidence and this report

Technical sources: [R3] TDK MPU6050 register map; [R4] Bosch BMP280 datasheet; [R5] Nooploop TOFSense-F hardware/protocol; [R7] Ultralytics detection; [R8] MIT OCW 16.30 feedback control; [R9] ROS REP-105 frames; [R10] Raspberry Pi RP2350/RP2040 specifications; [R11] TI CC1101; [R12] Semtech LoRa; [R13] ROS 2 actions/QoS. Full primary-source URLs are preserved in reference/pico-system-edition-v4.1.md.

Project evidence: [P1] secured-mount calibration; [P2] gyro restart; [P3] FIFO/gyro frame and timing; [P4] offline attitude replay and adverse models; [P5] software input guards. evidence-manifest.json preserves paths/hashes. [P6] inspected Pico/ROS main commit bce665ad97adbfcb282f02970a1db599e67e6479; exact files in source-inventory.json. Source inspection is not hardware acceptance.

Supporting dossier: 38-page prior edition; full materials CSV; requirements register; source/evidence manifests; figure assumptions; preserved failures. Private originals require a sanitized examiner evidence package; public hashes alone do not replace logs.

Provenance: green diagrams and analytic plots are labelled; historical replay remains unchanged. Cover = AI F450 concept, not a hardware photograph. Edition dated 27 October 2026; later review dates stay in provenance. No firmware, calibration, motor authorization or flight acceptance is changed by this editorial edition.